KV-Cache Scaling

Predicting the context limit before reaching it

Integrated scientific implementation and evidence report | 19 October 2026

Status: RESEARCH CRITERIA PENDING

The analysis, plots and transfer calculations are integrated and reproducible. Verified coverage: 52 primary probes across 9 measured lengths (512-114688 tokens). Requirement status below distinguishes completed measurements from unmet scientific criteria.

Built capability	Evidence / limit
Memory model	Unique tied weights and GQA-aware KV equation; original numerical prediction preserved
GPU observations	52 primary probes across 9 measured lengths (512-114688 tokens)
Recovery	External checkpoint restoration passed after both Colab copies were removed
Analysis	Validated immutable capsules; regressions require multiple lengths; unexplained bytes remain explicit
Transfer	Four config-only model/GPU predictions; conditional published KV-only comparison

The project followed five phases: derivation and preregistration, isolated probes, checkpointing and safety, empirical analysis, and transfer with reproducible presentation. Drive remained disconnected; review provenance is preserved in the original frozen records.

Unmet evidence: Paired prospective held-out validation at L=114688. Actual OOM boundary unobserved; conservative safe-budget censoring is not an OOM threshold.

1. Model, derivation and committed prediction

Primary model: Qwen/Qwen2.5-1.5B, FP16, batch size 1. Sequence length L is one cold prefill of L tokens with an empty DynamicCache and logits_to_keep=1. HF SDPA is forced to the efficient backend. No generation-to-L, eager fallback or FlashAttention-2 is used.

Symbol	Meaning	Value
N	Transformer layers	28
H / I	Hidden / intermediate width	1536 / 8960
Q / K / D	Query heads / KV heads / head dimension	12 / 2 / 128
V	Vocabulary	151936
s	Bytes per FP16 cache element	2

q=QD; k=KD. Unique parameters P = VH + N[Hq + 2Hk + qH + q + 2k + 3HI + 2H] + H. The tied input/output embedding is counted once.

W=2P=3087428608 bytes. KV(L,B)=2BLNKDs=28672BL bytes. Using Q in place of K overestimates KV by 6x; duplicating the tied embedding adds 466747392 bytes.

Advance budget model: M(L,B)=W+BL(28672+a)+H0. The central assumption is a=78336 bytes/token and H0=536870912 bytes. L(U,B)=max(0,floor((U-W-H0)/(B(28672+a)))). U is usable memory after context, before weights; context must not be subtracted again.

Original T4 usable budget: 15527116800 bytes. Central estimate: 111232 tokens; assumption range: 76024-131072 after the config cap. This is not a measured confidence interval or an OOM observation. The model-card quality limit is 32768; larger lengths are synthetic memory stress.

Prediction commit 7baff5121ed643f27ba06959065855a8913f8e8c and later receipt 976158c82ccb10f12817af2ee16b76ade13f0cf3 preceded model work. Full package inventories, the GPU UUID and source fingerprints remain in raw evidence.

2. What the saved GPU data establishes

Allocator	L	n	Peak allocated bytes	Peak reserved bytes	Cache bytes
default	512	3	3146681856	3330277376	14680064
default	1024	3	3195448832	3403677696	29360128
default	2048	3	3295079936	3529506816	58720256
default	4096	3	3487002112	3693084672	117440512
default	8192	3	3877137920	4240441344	234881024
default	16384	3	4657409536	5171576832	469762048
default	32768	3	6217952768	6731857920	939524096
default	65536	3	9339039232	10171187200	1879048192
default	114688	3	14020668928	15319695360	3288334336
expandable	512	3	3144715776	3164602368	14680064
expandable	1024	3	3193482752	3202351104	29360128
expandable	2048	3	3291016704	3317694464	58720256
expandable	4096	3	3486084608	3529506816	117440512
expandable	8192	3	3876220416	3925868544	234881024
expandable	16384	3	4656492032	4764729344	469762048
expandable	32768	3	6217035264	6400507904	939524096
expandable	65536	3	9338121728	9695133696	1879048192
expandable	114688	1	14019751424	14644412416	3288334336

Repeated points use fresh worker processes. The cache contains 14680064 bytes at L=512 and reduced logits are FP16 (1,1,151936). Per-length medians and ranges come from the recorded repeats. Kernel traces remain separate diagnostics.

Retained residual fraction: 0.133456% to 0.303312% of retained allocation across primary points.

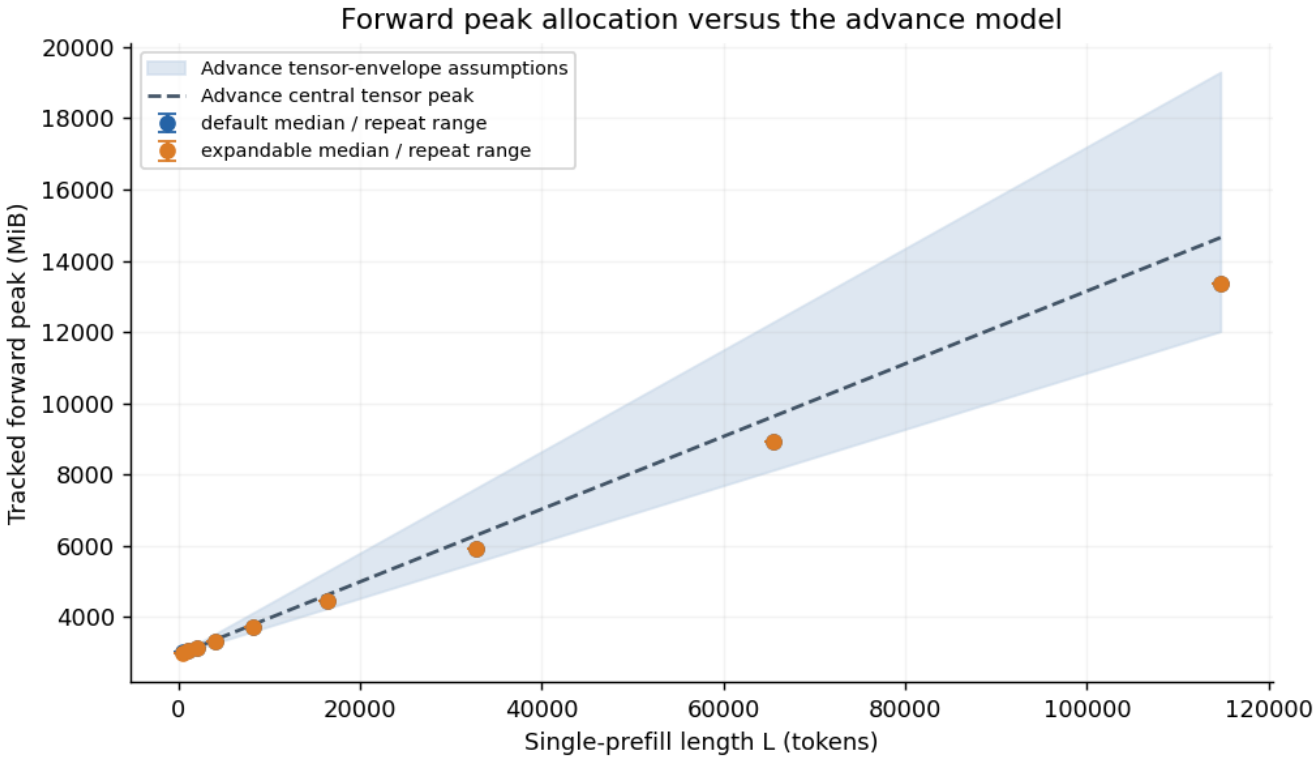
All snapshots respect reserved >= allocated and forward peaks cover synchronized post-forward allocation. Retained-minus-released allocation matches inspected cache bytes. Driver samples are sampled observations, not exact driver high-water marks.

Both allocators cover the five required lengths with three repeats.

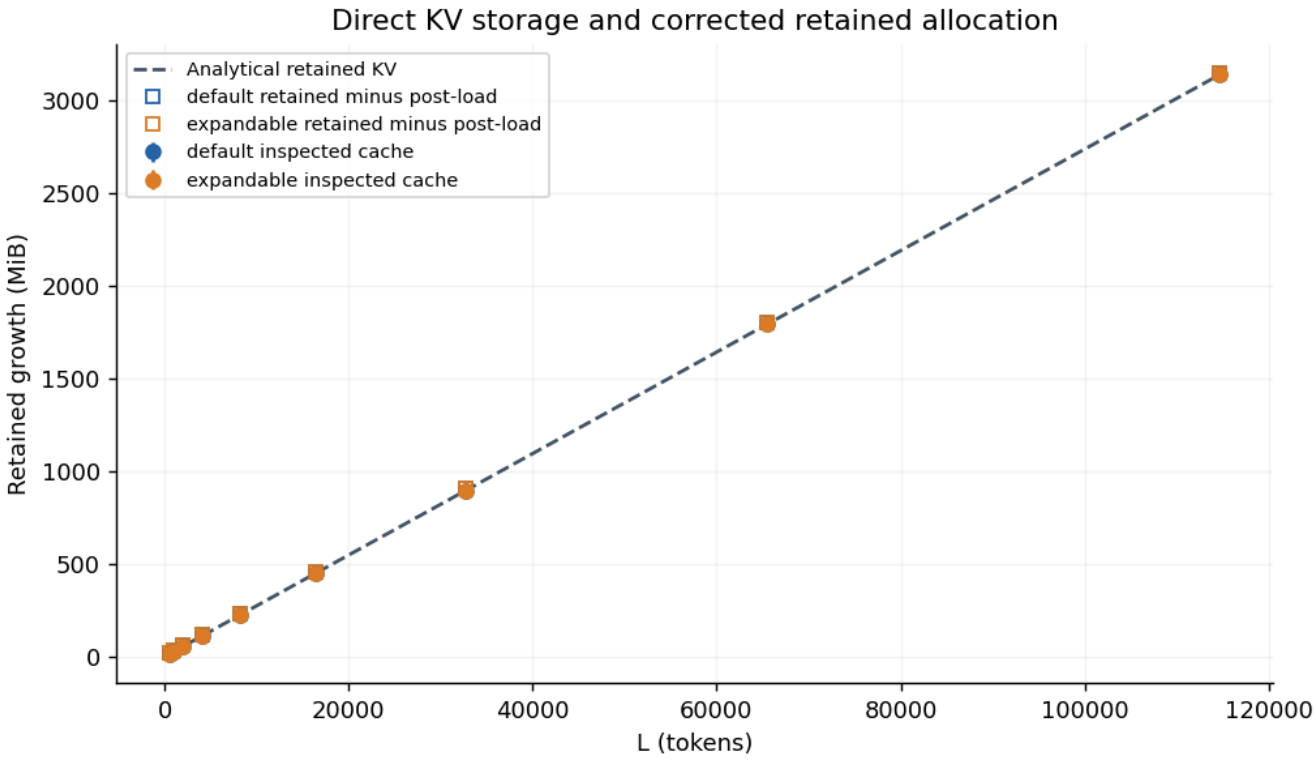
default: directly inspected KV slope = 28672.000000000004 bytes/token; R-squared = 1.0. Total-peak fits remain separate.

expandable: directly inspected KV slope = 28672.000000000004 bytes/token; R-squared = 1.0. Total-peak fits remain separate.

3. Memory curves and retained KV

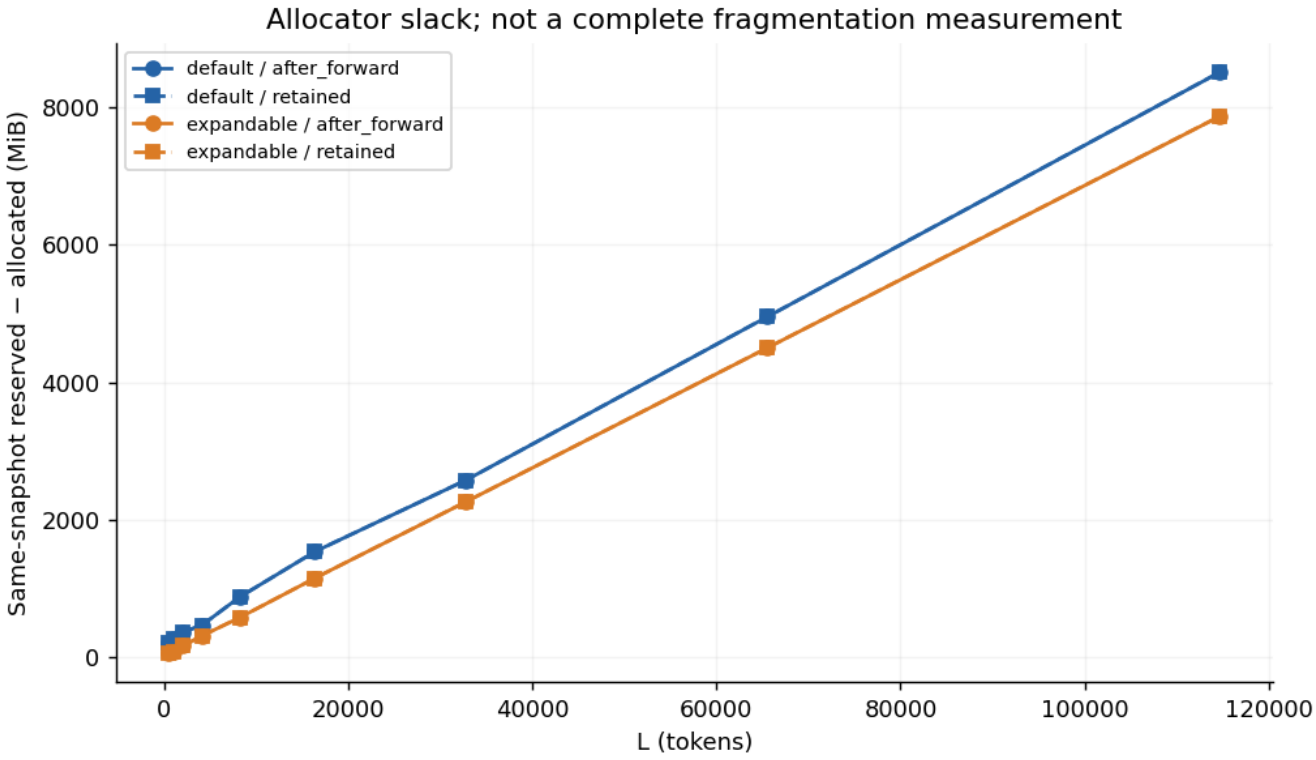


The band is an advance tensor-envelope assumption, not measured uncertainty. Points are observations; the unmeasured curve remains visibly labelled.



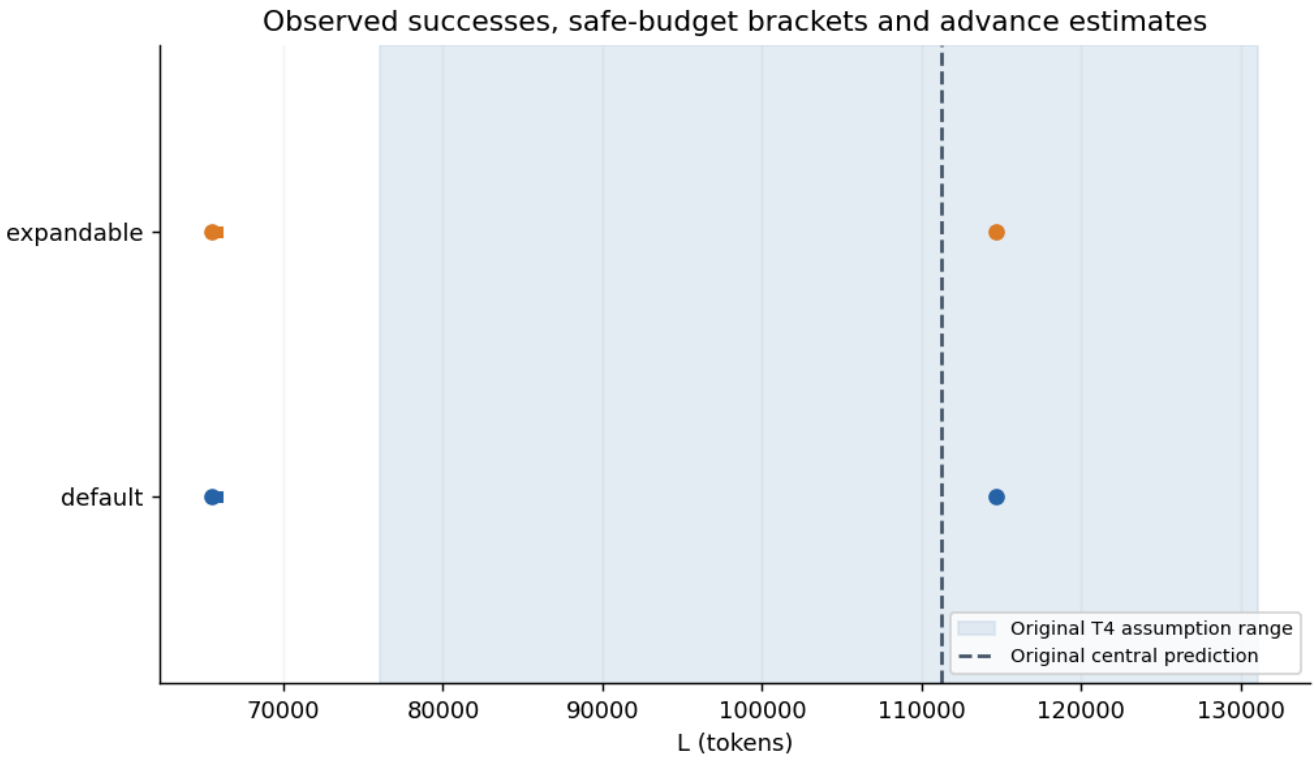
Direct cache and corrected retained allocation are separate quantities. Fits require multiple distinct lengths.

4. Allocator slack and boundary status



Same-snapshot reserved minus allocated is allocator slack. It is not, by itself, fragmentation.

4. Boundary evidence (continued)



Observed success, safe-budget bracket, actual OOM upper bound and advance prediction remain distinct. No hardware threshold is fabricated.

Expandable-minus-default conservative safe-limit interval: -256 to 256 tokens; observed-success delta 0. A common analytical admission ceiling can censor both allocators; this is not a hardware-limit delta.

default: observed success 114688 tokens; OOM upper point None; original prediction error interval None percent. The interval assumes monotone feasibility and does not identify an exact threshold.

expandable: observed success 114688 tokens; OOM upper point None; original prediction error interval None percent. The interval assumes monotone feasibility and does not identify an exact threshold.

Held-out default L=114688, n=3: frozen post-sweep fit predicted 14019203874 bytes; observed median 14020668928; error -0.0104%. This is a later validation, not a replacement advance prediction.

Held-out expandable L=114688, n=1: frozen post-sweep fit predicted 14019751424 bytes; observed median 14019751424; error 0.0000%. This is a later validation, not a replacement advance prediction.

5. Signed prediction discrepancy

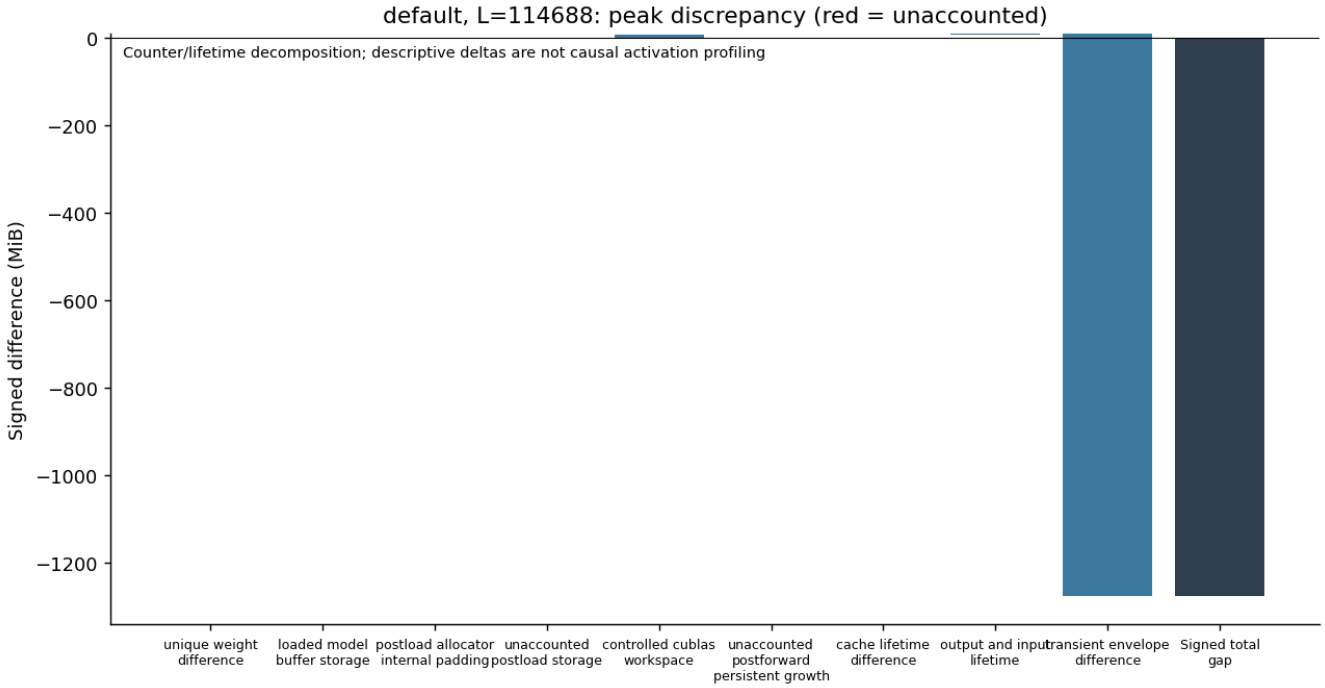
Allocator / L / repeats	Median gap bytes	Max closure bytes	Max unaccounted bytes	All repeats
default / 512 (n=3)	4465152	0	0	PASS
default / 1024 (n=3)	-1555968	0	0	PASS
default / 2048 (n=3)	-11501056	0	0	PASS
default / 4096 (n=3)	-38731264	0	0	PASS
default / 8192 (n=3)	-86900224	0	0	PASS
default / 16384 (n=3)	-183238144	0	0	PASS
default / 32768 (n=3)	-375913984	0	0	PASS
default / 65536 (n=3)	-761265664	0	0	PASS
default / 114688 (n=3)	-1339293184	0	0	PASS
expandable / 512 (n=3)	2499072	0	0	PASS
expandable / 1024 (n=3)	-3522048	0	0	PASS
expandable / 2048 (n=3)	-15564288	0	0	PASS
expandable / 4096 (n=3)	-39648768	0	0	PASS
expandable / 8192 (n=3)	-87817728	0	0	PASS
expandable / 16384 (n=3)	-184155648	0	0	PASS
expandable / 32768 (n=3)	-376831488	0	0	PASS
expandable / 65536 (n=3)	-762183168	0	0	PASS
expandable / 114688 (n=1)	-1340210688	0	0	PASS

Summary uses median signed gaps and worst-case absolute closure/unaccounted bytes across repeats. Every repeat-level row is preserved in the accompanying Markdown report and discrepancy CSV.

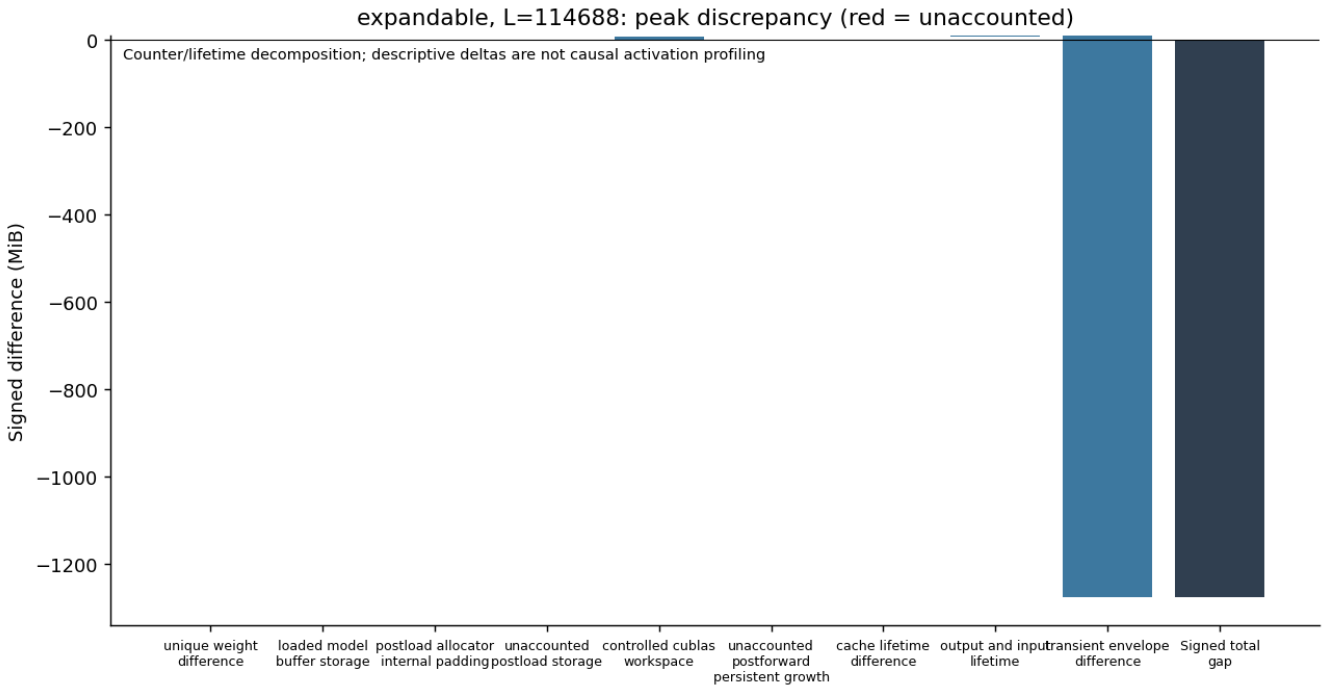
The signed identity telescopes across post-load, after-forward, retained and released counters. Arithmetic closure is separate from explaining causes. Unknown baseline storage and persistent post-forward growth are retained as unaccounted, even if a plausible source-level hypothesis exists.

For a material gap (at least 10 MiB), supported attribution must reach 90% under both gap and absolute-contribution denominators. Smaller gaps require at most 1 MiB unaccounted and an ill-conditioned-ratio label. Computed coverage status: True.

5. Discrepancy components (continued)



Default allocator. Red components remain unaccounted. Transient counter deltas do not establish pure activation memory.



Expandable allocator. The same signed accounting and explanatory-coverage rules apply.

Separate diagnostic comparisons

Mode	Allocator	L	Repeats	Peak delta bytes
chunked_dynamic	default	512	3	1048576
chunked_dynamic	default	2048	3	-100139008
chunked_dynamic	default	8192	3	-465567744
chunked_dynamic	default	32768	3	-1900544000
static_cache	default	512	3	524288
static_cache	default	2048	3	8388608
static_cache	default	8192	3	134217728
workspace_control	default	512	3	-8519680
workspace_control	expandable	512	3	-8519680

Diagnostic minus matching primary median. Chunking and static caching change execution semantics. Workspace interventions change GEMM workspace policy. All are excluded from the primary cache slope and hardware-boundary observations.

Pinned HF 4.51.3 creates an explicit query-by-capacity causal mask for StaticCache. A full-prefill mask is L by L even with efficient SDPA; chunking bounds the query dimension. Peak deltas include this mask behavior, not solely cache storage.

PyTorch 2.6 source allocates 8519680 default cuBLAS workspace bytes on a non-Hopper CUDA GPU. Three matched same-allocator zero-workspace controls directly support L=512 rows. Other lengths use labelled source-supported fixed-size inference only when their persistent delta matches exactly.

6. Transfer to two unrun models and GPUs

Model	GPU	Assumed usable GiB	Raw central L	Capped L	Capped range
Qwen2.5-3B	A100-40GB	37/38/39	251278	32768	32768-32768
Qwen2.5-3B	RTX-4090-24GB	22/23/23.5	132571	32768	32768-32768
Qwen2.5-7B	A100-40GB	37/38/39	109629	109629	72387-131072
Qwen2.5-7B	RTX-4090-24GB	22/23/23.5	39097	39097	22613-55262

Only official config files were downloaded. No 3B/7B checkpoint was loaded and neither target GPU was measured. VRAM budgets and activation envelopes are explicit assumptions. The 3B config cap censors its memory estimate; censoring is not an observed OOM.

Matching KV architecture	Predicted block bytes	Shown MB	Error if MiB	Error if decimal MB
Qwen2.5-3B	589824	0.56	0.4464%	5.3257%
Qwen2.5-7B	917504	0.87	0.5747%	5.4602%

The vLLM team publishes physical 16-token KV blocks across all layers. Base/Instruct KV fields agree in pinned configs. Under a two-byte cache assumption the comparison checks retained KV scaling only. The source labels MB without defining decimal versus binary units, so both displayed-value errors are reported.

The published serving engine has different attention, allocation and scheduling semantics. This evidence does not validate the transferred HF single-prefill peak thresholds, activation envelopes or target usable VRAM.

7. Acceptance status and reproducibility

ID	Status	Criterion
FR1	PASS	Pinned config, KV heads and tied-weight arithmetic
FR2	PASS_FOR_SAVED_ACCEPTANCE	Exact weights/cache verified; residuals reported separately
FR3	PASS_FOR_SAVED_ACCEPTANCE	Original prediction 7baff51 and later receipt 976158c precede model work
FR4	CHECKPOINTED_GRID_PASS	Fresh workers and per-probe external checkpoint barriers; original acceptance restoration verified
FR5	PASS	Five lengths, three repeats and retained-cache slope required
FR6	SAFE_BUDGET_CENSORED	One-use OOM bracket and paired safe-budget delta are distinct; an exact hardware-limit delta is not established
FR7	PASS	Closed signed table; unsupported bytes remain unaccounted
FR8	CONDITIONAL_KV_COMPARISON	Four unrun transfer cells; published KV block comparison with dtype/unit caveats

Regeneration is CPU-only: install the package with requirements-analysis.txt, run kvcache analyze on the immutable capsule, run kvcache generalize, and invoke scripts/build_report.py. Every output directory must be new. Full command lines appear in report.md and the reproduction guide.

Historical acceptance preserved an import-time torchao preflight failure and verified restoration after deleting both Colab checkpoint copies. Removing unused torchao restored cold imports. That bounded acceptance consumed no OOM allowance; full cloud-session restoration of the same GPU identity is not claimed.

The requirement table is computed from saved evidence. Safe-budget boundaries remain distinct from actual hardware OOM thresholds. Unsupported causes remain unaccounted. The original prediction is preserved; the prospective implementation amendment and later receipt record the changed GPU/source before extended probes.

Primary sources

Executed audit notebook

Official vLLM KV block table

PyTorch v2.6.0 workspace source

Pinned HF 4.51.3 Qwen2 masking source

Official transfer config URLs/revisions/hashes: configs/transfer/provenance.json. Source capsule hashes, analysis code hashes and report package versions are in provenance and manifest files.